

We have focused on the **phases** so far. What about the boundaries between the phases

depicted in Fig. (5.1) and Fig. (5.2). There the two adjacent phases are in equilibrium so can coexist. At the triple point all three phases are in equilibrium.

Let us focus on the simpler cases of the lefthand two diagrams in Fig. (5.1).

5.1 Van der Waals fluid

We will firstly use a model, the [van der Waals](#) equation of state, which does not capture the solid phase, but describes the gas and liquid, providing a critical point C , as in Fig. (5.1).

Consider the equation of state for one mole of gas

$$\left(p + \frac{a}{V^2}\right)(V - b) = RT \Leftrightarrow f(T, p, V) = 0, \quad (5.1)$$

where a and b are constants. If the terms in [red](#) were zero, then the equation of state is that of an ideal gas.

The physics of the two terms becomes clearer if we solve for p :

$$p = -\frac{a}{V^2} + \frac{RT}{V - b}. \quad (5.2)$$

Note as $V \rightarrow b$ $p \rightarrow \infty$. b represents a **minimum** volume for one mole of gas, resulting (microscopically) from a hard core like the $1/r^{12}$ term in Lenard-Jones from Y1 Temperature and Matter. The first term on the RHS of Eq. (5.2) denotes a **decrease** of the pressure compared to an ideal gas. Microscopically, it represents the attractive tail in the interaction between the atoms – like the van der Waals interaction in Lenard-Jones potential. As the atoms approach the container wall they are moving away from the attractive potential of the gas "behind them" and hence slow down, so their deposition of momentum on the container wall is reduced and hence the pressure reduced.

What form do the isotherms take for this equation of state? We can see that at large V

$$p \sim \frac{RT}{V},$$

as in the ideal gas. And $p \rightarrow \infty$ as $V \rightarrow b$. Between those two behaviours, either $p(V, T)$ is featureless as in the lefthand figure Fig. (5.3), or has (at least) one minimum and maximum as in the righthand figure in Fig. (5.3). To go between these possibilities, there is a critical value of the parameters where the maximum and minimum merge into a point of inflection – as in the [green](#) curve in Fig. (5.3).

Since the equation is not a quadratic, it is not easy to find the general solution, but we may determine the critical point (with the point of inflection). There, both

$$\left(\frac{\partial p}{\partial V}\right)_T = 0 \quad \text{and} \quad \left(\frac{\partial^2 p}{\partial V^2}\right)_T = 0.$$

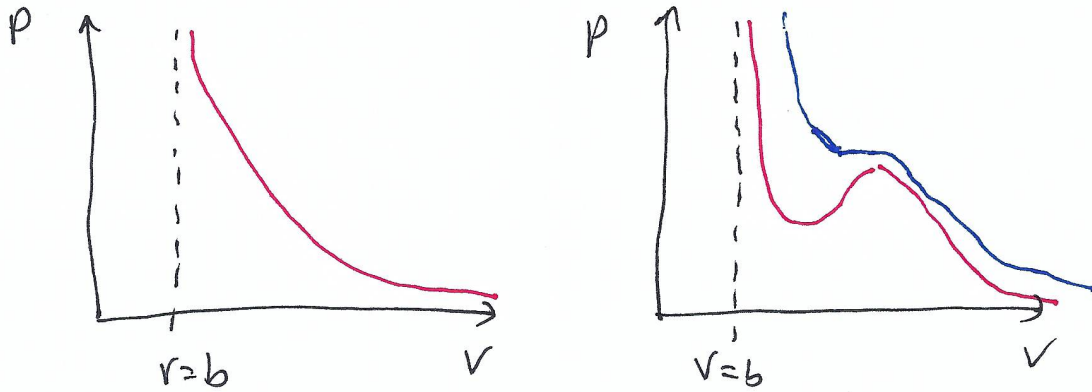


Figure 5.3: Left: A featureless isotherm, apart from divergence at the minimum volume, b . Right: the other possibility, a maximum and minimum in the red curve and a point of inflection for the critical case in the green curve.

We see, for a point of inflection:

$$\left(\frac{\partial p}{\partial V} \right)_T = -\frac{RT}{(V-b)^2} + \frac{2a}{V^3} = 0 \quad (5.3)$$

$$\text{and} \quad \left(\frac{\partial^2 p}{\partial V^2} \right)_T = \frac{2RT}{(V-b)^3} - \frac{6a}{V^4} = 0. \quad (5.4)$$

Solving for RT from both Eq. (5.3) and Eq. (5.4) and equating we find:

$$\begin{aligned} RT &= \frac{2a(V-b)^2}{V^3} = \frac{3a(V-b)^3}{V^4} \\ \Rightarrow \quad \frac{3(V-b)}{V} &= 2 \\ \Rightarrow \quad V_c &= 3b. \end{aligned} \quad (5.5)$$

Substituting Eq. (5.5) into Eq. (5.4) yields

$$T_c = \frac{8a}{27Rb}. \quad (5.6)$$

Finally substituting both Eq. (5.5) and Eq. (5.6) into the equation of state, Eq. (5.2), we see

$$p_c = \frac{a}{27b^2}. \quad (5.7)$$

At the critical temperature, T_c , the isothermal compressibility, κ_T diverges:

$$\kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T \underset{T \rightarrow T_c}{\sim} \infty \quad (5.8)$$

i.e. the material becomes infinitely compressible.

Worse, for $T < T_c$, $\kappa_T < 0$ (examine the righthand graph of Fig. (5.3)) – the material expands if one applies pressure, it is unstable. Notice the gradient, $(\partial V / \partial p)_T$, in the righthand figure in Fig. (5.3) is positive over a region of V (remember the minus sign in the definition of κ_T ,

Eq. (5.8)). The resolution is not immediate, but is that the material separates into a liquid region and a gas region at the same pressure. As the volume is decreased, the proportion of gas drops and the proportion of liquid grows – as the latter is denser until there is nothing but liquid. For smaller volumes there is no ambiguity, the material is compressed as a liquid with positive κ_T (and similarly for the region with $\kappa_T > 0$, at high temperature, where there is a stable gas).

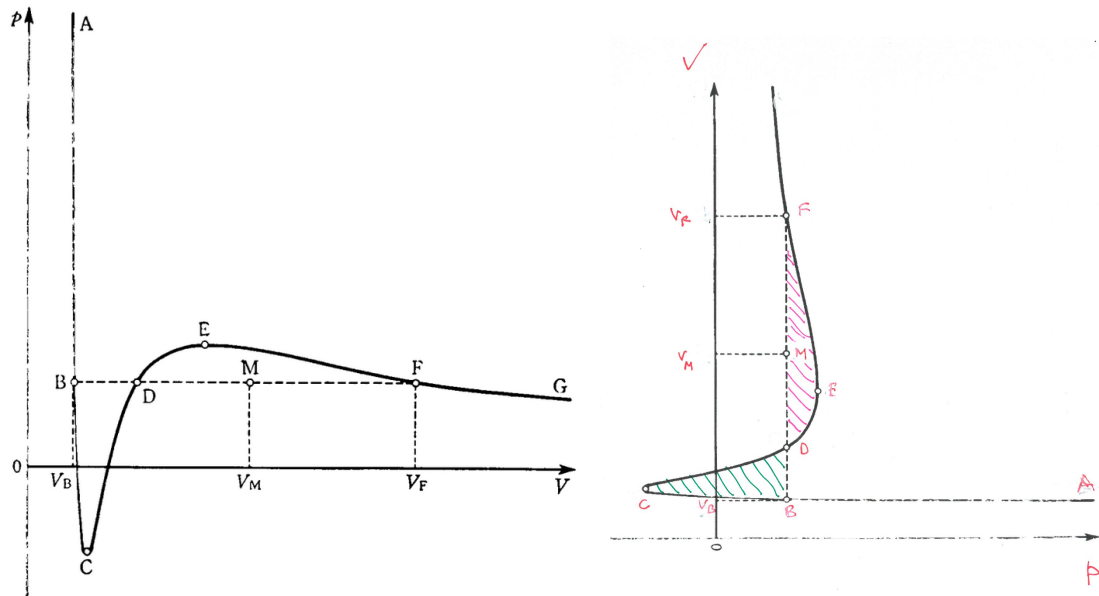


Figure 5.4: Left: a van der Waals isotherm and the notation for the Maxwell construction. Credit: Kubo, R. *Thermodynamics*, Elsevier (1968). Right: The green and pink regions of integration for the Gibbs free energy, with the axes reflected and rotated for clarity.

Let us now analyse this in more detail. Consider Fig. (5.4) depicting a van der Waals isotherm. Notice that the region between C and E on the isotherm is unstable as $\kappa_T < 0$. The resolution, as mentioned above, is that there is a mixture of the two phases (in reality usually separated spatially by gravity ensuring the less-dense phase resides on top of the denser one). The two phases must be at the same pressure – otherwise they will not be in mechanical equilibrium. An example of the two phases having the same pressure are the two points B and F .

At a fixed pressure the Gibbs free energy will be a minimum. Let us calculate the free energy difference $G(B) - G(F)$, and determine under what conditions this difference is zero – so the two phases may be in equilibrium with each other.

We are considering an isotherm $dT = 0$ and hence $dG = Vdp$, thus

$$\begin{aligned}
 \text{(Green area)} &= \int_C^D V dp - \int_C^B V dp \\
 &= [G(D) - G(C)] - [G(B) - G(C)] \\
 &= G(D) - G(B) \\
 \Rightarrow G(D) - G(B) &= \text{Area}(DCBD) .
 \end{aligned} \tag{5.9}$$

$$\begin{aligned}
 \text{and (Pink area)} &= \int_F^E V dp - \int_D^E V dp \\
 &= [G(E) - G(F)] - [G(E) - G(D)] \\
 &= G(D) - G(F) \\
 \Rightarrow G(D) - G(F) &= \text{Area}(FEDF) .
 \end{aligned} \tag{5.10}$$

Subtracting Eq. (5.9) from Eq. (5.10) we see:

$$G(B) - G(F) = \text{Area}(FEDF) - \text{Area}(DCBD) \tag{5.11}$$

If $G(B) > G(F)$, then atoms would evaporate, to lower the free energy. And would condense from the vapour if $G(B) < G(F)$. So for stability of the mixture, the free energies must be equal, implying that

$$\text{Area}(FEDF) = \text{Area}(DCBD) , \tag{5.12}$$

the celebrated "equal area" Maxwell construction, selecting the correct horizontal isotherm. How do we determine the relative fraction of liquid, x , and vapour, $1 - x$, at a point corresponding to V_M ? The volumes of liquid and vapour must add up to V_M :

$$\begin{aligned}
 xV_B + (1 - x)V_F &= V_M \\
 \Rightarrow xV_B + (1 - x)V_F &= xV_M + (1 - x)V_M \\
 \Rightarrow \text{Ratio of mols of liquid to solid: } \frac{x}{1 - x} &= \frac{V_F - V_M}{V_M - V_B} .
 \end{aligned}$$

What do the regions $B-C$ and $E-F$ signify? They are apparently stable as $\kappa_T > 0$, but the Maxwell construction implies those regions are not stable. These regions correspond to *metastable* liquid ("superheated liquid") and metastable vapour ("supercooled vapour"). The calculation of their lifetime is a calculation in statistical mechanics which is hard to perform in practice as the result (and experiments) are very sensitive to the purity of the gas and the nature of the container (both impurities and the container surface may nucleate the equilibrium phase).

There is a caveat to the above analysis – we appear to be taking the unstable regions seriously in the paths of integration over p . This can be fixed – with considerable difficulty, see [Kubo, R. Thermodynamics, p. 284-288, Elsevier \(1968\)](#).

Turning to the case of a material with all three phases, it is hard to provide a simple model which encompasses all of the phases, but we may portray the outcome in Fig. (5.5). Note on

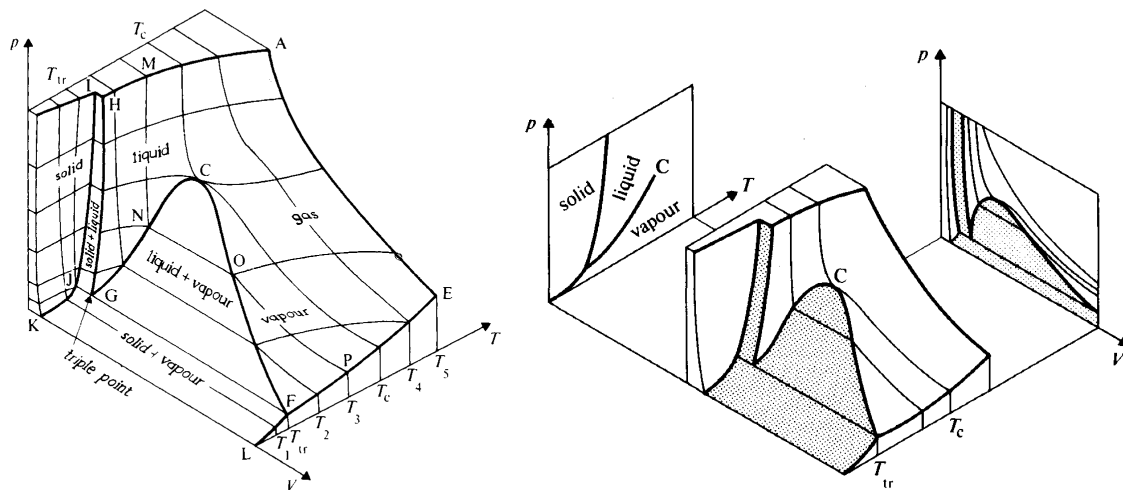


Figure 5.5: Left: the surface representing the equation of state in the p - V - T space. Right: the projections of the lines of coexistence into the (left) p - T plane and (right) the p - V plane. On the latter there are isotherms away from the critical point indicated. In principle one could depict lines of constant volume on the former. This is not usually done. Credit: Adkins, p 122 and 123.

the left we represent the surface from the equation of state, $f(T, p, V) = 0$ (see Eq. (5.1)), in the three-dimensional p - V - T space. We may project this surface onto the two planes. p - T and p - V , indicated in the righthand figure of Fig. (5.5).

Let us compare two paths cooling down vapour on the two projections depicted in the righthand image of Fig. (5.5). The paths are depicted in Fig. (5.6). The path at constant pressure is quite straightforward in both projections. The path at constant volume is more interesting: it is a vertical line in the p - V plane, which shows (for this value of V) that the path travels through the two mixed-phase regions, vapour-liquid at higher p and vapour-solid at lower p . In the p - T plane this means once it reaches the liquid-vapour mixed phase region it tracks down that phase boundary and then the vapour-solid one.

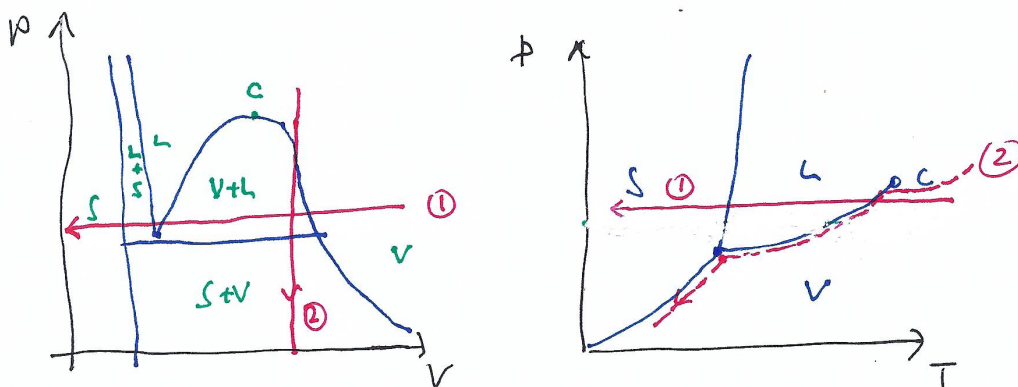


Figure 5.6: Left: the p - V cross section through the equilibrium surface $f(p, V, T) = 0$. Right: the p - T cross section through $f(p, V, T) = 0$. The paths ①, at constant p and ② at constant V .

5.2 Latent heat and the Clapeyron equation

A change of phase, apart from at the critical point, is accompanied by a spike in the heat capacity, the *latent heat*. The latent heat, L , is required to melt a solid (latent heat of melting) or vapourise a liquid (latent heat of vapourisation) at constant temperature and pressure. Unfortunately, the *specific* latent heat, per mole and per kg, are also denoted by L . In the projection on the p - V projection, the latent heat is provided over the horizontal part of the isotherm; in the p - T projection the heat is provided at a point on the phase boundary, see Fig. (5.7).

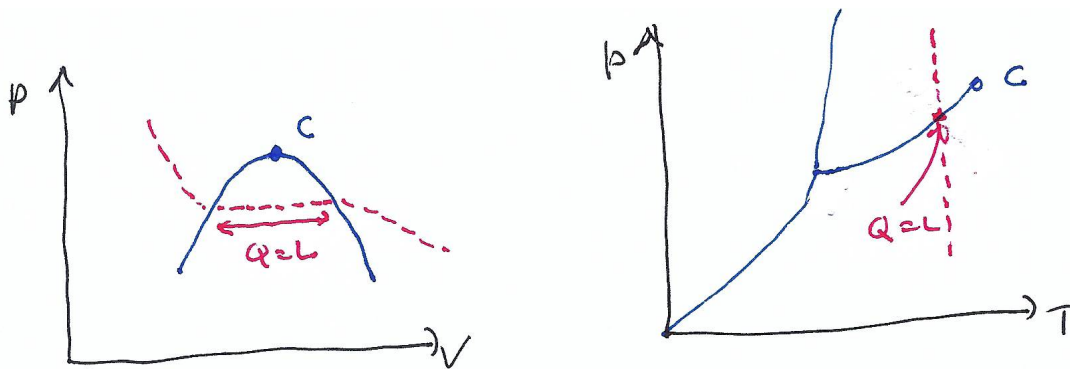


Figure 5.7: Left: An isotherm in the p - V plane, crossing the liquid-vapour mixed-phase region, indicating the latent heat, L , being absorbed as the region is traversed. Right: the same isotherm in the p - T plane. Here the latent heat is associated with one point on the plane.

The *Clapeyron* equation (actually discovered by Carnot, rediscovered by Clapeyron and Clausius and carefully studied experimentally by Kelvin and his brother, James Thomson) is a differential equation for the phase boundary $p(T)$, in the p - T plane. Consider two phases of a substance,

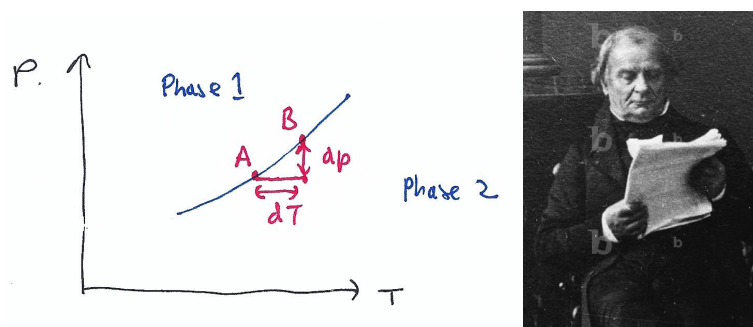


Figure 5.8: Left: the geometry for the Clapeyron equation in the p - T plane: two nearby points, A and B , on a phase boundary, and the changes in dp and dT . Right: Émile Clapeyron.

with masses m_1 and m_2 , and total mass $m_t = m_1 + m_2$. The system is closed. The specific Gibbs free energies (Gibbs energy per unit mass) are $g_1(p, T)$ and $g_2(p, T)$. The total Gibbs free energy is

$$G = m_1 g_1 + m_2 g_2 ,$$

minimised at equilibrium, with constant T and p . So

$$\begin{aligned}
 dG &= 0 \quad (\text{And } dg_i = 0.) \\
 \Rightarrow \quad dm_1 g_1 + dm_2 g_2 &= 0 \\
 \Rightarrow \quad dm_1 g_1 - dm_1 g_2 &= 0 \quad (\text{As } dm_t = 0.) \\
 \Rightarrow \quad dm_1 (g_1 - g_2) &= 0 \\
 \Rightarrow \quad g_1 &= g_2 .
 \end{aligned}$$

Now consider two nearby points on the phase boundary, $p(T)$, see Fig. (5.8). Then in terms of the separation of the two points by dT and dp , we see, using s_i for the specific entropies and v_i for the specific volumes,

$$\begin{aligned}
 dg_1 &= -s_1 dT + v_1 dp \\
 dg_2 &= -s_2 dT + v_2 dp ,
 \end{aligned}$$

and since $g_1 = g_2$, $dg_1 = dg_2$, hence

$$\begin{aligned}
 -s_1 dT + v_1 dp &= -s_2 dT + v_2 dp \\
 \Rightarrow \quad \frac{dp}{dT} &= \frac{\Delta s}{\Delta v} = \frac{\Delta S}{\Delta V} .
 \end{aligned}$$

Now the change of entropy is obtained from

$$\Delta S = \frac{Q}{T} = \frac{L}{T} . \quad (5.13)$$

Hence

$$\frac{dp}{dT} = \frac{L}{T \Delta V} , \quad (5.14)$$

the *Clapeyron* equation. Notice it is associated with a point on the phase boundary as the volume change and latent heat are both dependent on the position on the phase boundary.

5.3 The Third law

What is the value of the entropy of a system? So far we have only calculated the *change* in entropy between initial and final states. Quantum mechanics is needed to answer this question in a satisfactory manner.

But considering a Carnot engine acting as a refrigerator allows some insight, see Fig. (5.9). The efficiency is

$$\eta = \frac{Q_1}{W} = \frac{Q_1}{Q_2 - Q_1} = \frac{T_1}{T_2 - T_1} \rightarrow 0 \quad \text{when } T \rightarrow 0 . \quad (5.15)$$

So removing heat becomes very inefficient as the temperature of the cold reservoir tends to 0K. One statement of the Third Law is that it is impossible to reach zero temperature in a finite number of steps.

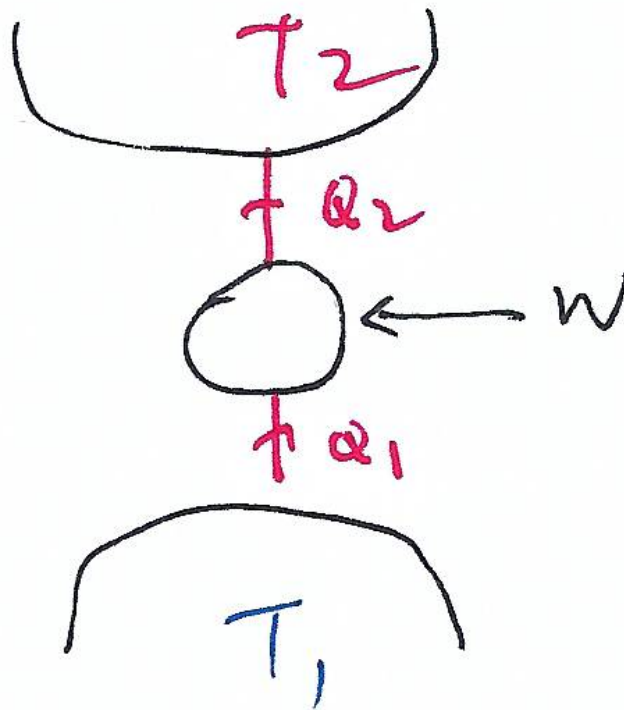


Figure 5.9: A Carnot engine removing Q_1 from the low temperature body and depositing Q_2 at the high temperature reservoir, performing work W to achieve this.

We may also determine one property of the specific heat. Noting that

$$\Delta S = \int_{T_0}^0 \frac{C dT}{T}$$

diverges unless $C \rightarrow 0$. Since the entropy is finite even at $T = 0$, we deduce that $C \rightarrow 0$ as $T \rightarrow 0$. This may be interpreted using quantum mechanics and statistical physics.